Choices for Embedded Computing (MCU/MPU)

Deterministic Control Systems

	Unit Cost (>5,000 Units)	Package Size	Number of IO Pins	Have Development Unit	Working Example Firmware
TI MSP430 5-Series	$2.25	5x5	47	Y	Y
TI MSP430 2-Series	$2.65	2.8x2.8	32	Y	Y
TI Tiva	$10.80	10x10	140	Y	Y
SiLabs EFM32	$1.36	6x6	24	Y	Y
Atmel Atmega	$3.22	5x5	32	Y	Y
Microchip PIC32	$8.08	16x16	120	Y	N
NXP Kinetis	$2.20	5x5	23	Y	Y

USB

TI Tiva	$10.80	10x10	140	Y	Y
SiLabs EFM32	$1.36	6x6	24	Y	Y
ST Micro STM32	0.57	3x3	24	Y	Y

Linux-Realtime

TI Sitara	$7.12	13x13	200	Y	Y
Qualcom Snapdragon	-	-	-	Y	Y

Linux-Desktop

TI Sitara	$7.12	13x13	200	Y	Y
Amlogic s905	-	-	-	Y	Y

Smallest Form Factor

Atmel ATTINY 85	$0.85	1.6x1.4	10	Y	Y
NXP RS08	$0.50	3x3	2	Y	N
NXP Kinetis KL03	$1.16	2x1.6	14	N	N

Lowest Power

TI MSP432	$3.22	14x14	84	Y	Y
NXP Kinetis KL81	$3.91	5x5	60	N	N

Ethernet

	Unit Cost (>5,000 Units)	Package Size	Number of IO Pins	Have Development Unit	Working Example Firmware
TI Tiva	$10.80	10x10	140	Y	Y
Cypress S6E2CCA	$9.26	12x12	190	Y	Y

Wi-Fi

TI CC3200	$6.93	9x9	27	Y	Y
SiLabs WGM110	$13.11	16x24	0	Y	N
Cypress WICED	$19.94	-	-	Y	N
Marvell 88MC200	$13.85	8x8	80	Y	N

Bluetooth

TI CC2650	$3.75	4x4	15	Y	Y
NXP Kinetis-W	$2.88	7x7	32	N	N
Nordic NRF52	$2.52	6x6	32	Y	Y

Custom RF and Radio

TI CC430	$3.26	9x9	44	N	N
TI CC1310	$3.25	4x4	30	N	N

9/7/2016 (r1.1)

Legend
Jmr Starting Selection
Yes
No

Embedded Computing Families (MCU/MPU)

Strong Favorites

Tiva	ARM Cortex, GPIO Count, Clock Speed
MSP430 5-Series	Determinisim, Low-level System Operation & Control, Power Consumption

Excited & No Unknowns

CC3200	Full Wi-Fi with ARM Cortex level system control
CC2650	Full BT, BLE, ZigBee with ARM Cortex level system control

Easy to Develop & Full Featured

Silicon Labs EFM8/32	wonderful processor architecture with clean demos and product web presentation (site & docs)
TI Sitara	Industry standard for embedded desktop linux
Cypress FM4	Easy interaction and peripheral control, wonderful documentation, web interface and examples

Strong Tech Support but Tricky with Slow Ramp-In

Atmel (ATTINY 85)	Atmel has horrid IDE, weak examples and simplified yet verbose documentation
NXP	Every mcu has a different IDE & toolchain, documentation is minimal and examples are not easily accessible

Painful & Slow, But Full Featured

Microchip (PIC32)	Microchip has quirky development tools with horrid IDE. This was not the case circa 2010 at all, and is strong suspect of recent business moves and as such is high risk & unknown
STMicro	It's all there, and it works but seems to always take an arm and a leg to find it. The difficulty here in ramp-in is surprising and a perceived risk of consistent delays by Justin

Unknown or New

All Linux Platforms	More time spent here needed, a newer dimension of embedded platforms for Justin
Marvell	Appears to be a solid and strong potential candidate for Phase One for Control and also Wi-Fi platforms, hands-on research and practice is needed
Infineon	Unknown Family, appears to have soft potential
Nuovoton	Unknown Family, appears to have soft potential
Maxim	Appears to only have value if pre-existing relationships or opportunities with Maxim are present
Renesas	Appears to only have value if pre-existing relationships or opportunities with Renesas are present

Closed Source & Hard Entry

Qualcomm	Unknown Family, appears to have minimal value and only given pre-existing relations of direct opportunities present
Analog Devices	Unknown Family, appears to have minimal value and only given pre-existing relations of direct opportunities present